Load-Aware A* Path Planning for Quadruped Robots

Day 25 of 30 · · Week 4: Results Review & Integration · Member 1 — Results Section Input

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1. Nature of Work Done

Day 26 completes the Discussion section of the paper. Day 25 produced sub-sections VI-A through VI-E, covering performance analysis, theoretical justification, stability benefits, and scalability. Today's four tasks extend the Discussion with the three remaining components that academic reviewers specifically scrutinise: literature context (how this work positions against prior art), novelty claims (what is original and why it matters), and a full limitations analysis (what the algorithm cannot do and why).

Task	Description	Paper Section	Status
Task 1	Continue Discussion section (VI-F: Literature Continuation)	VI-F — Broader literature comparison	Complete ✔
Task 2	Compare with literature	VI-F with detailed prior-art comparison tables	Complete ✔
Task 3	Explain novelty — formal novelty claims	VI-G — Novelty statement with differentiation analysis	Complete ✔
Task 4	Discuss limitations — extended analysis	VI-H — Extended limitations with mitigation paths	Complete ✔

PART A — SECTION VI DISCUSSION: CONTINUED (VI-F, VI-G, VI-H)

Continuation of the Discussion first draft begun on Day 25. These three sub-sections complete Section VI. Member 4 merges with Section V results before final paper assembly.

VI-F. Comparison with Prior Literature

Load-Aware A* does not exist in a vacuum. Stability-aware planning and legged robot navigation have received sustained research attention over the past two decades. Situating this work within the literature serves three purposes: it demonstrates scholarly awareness, it identifies the specific gap this work fills, and it provides reviewers with a basis for judging the contribution's significance. The literature comparison is organised into four streams: graph-search path planners for legged robots, stability constraint methods, sampling-based planners, and learning-based approaches.

VI-F.1 Stream 1: Graph-Search Path Planners for Legged Robots

The foundational work in this stream is Hart, Nilsson, and Raphael's original A* formulation (1968), which established the admissible heuristic framework that Load-Aware A* extends. Subsequent work adapting graph-search methods to legged robots has primarily focused on footstep planning and terrain traversability rather than payload-induced stability.

Reference (Representative)	Approach	Stability Handling	Payload Handling	Gap vs Load-Aware A*
Hart et al. (1968) — Standard A*	Admissible heuristic graph search; minimise step count	**NOT PRESENT: treats all free cells as equivalent**	**NOT PRESENT: no physical model**	Load-Aware A* extends Eq. 7 with continuous S(n) penalty while preserving admissibility
Zucker et al. (2011) — Footstep planning	A*-based footstep sequencing on 2D height maps; terrain cost heuristics	**PARTIAL: terrain slope used as traversal cost; no CoM model**	**NOT PRESENT: no payload mass modelled**	Load-Aware A* explicitly models payload CoM displacement via Eq. 4; not approximated from terrain
Chestnutt et al. (2005) — Humanoid navigation	Lattice A* with footstep reachability constraints; 3D environment	**PARTIAL: support polygon feasibility checked per step; binary**	**NOT PRESENT: static robot only**	Load-Aware A* uses continuous S(n) penalty rather than binary feasibility; avoids completeness loss
Gayle et al. (2009) — Multi-robot planning	Decoupled A* with obstacle avoidance; kinodynamic constraints	**NOT PRESENT: no stability term**	**NOT PRESENT**	Load-Aware A* integrates stability into global cost function; not a post-processing filter

Literature gap — Stream 1: No prior A*-based planner for legged robots incorporates a payload-mass-dependent CoM displacement model as a continuous additive cost in the global search objective. Existing work either ignores payload entirely or handles terrain stability through binary feasibility constraints that risk completeness failure.

VI-F.2 Stream 2: Stability Constraint Methods for Quadruped Robots

A significant body of work addresses quadruped stability through constraint-based approaches: defining stability regions and forcing paths to remain within them. These methods differ fundamentally from Load-Aware A* in their treatment of the stability-path trade-off.

Approach Type	Method Description	Stability Model	Payload Support	Key Limitation vs Load-Aware A*
Static Stability Margin (SSM)	Ensures CoM remains inside support polygon; minimum margin criterion (McGhee & Frank, 1968)	**DIRECT: M(n) > threshold as hard constraint**	**NOT PRESENT: static robot mass only**	Binary exclusion causes completeness failure when all paths cross low-M regions; no continuous gradient
Zero-Moment Point (ZMP)	Dynamic stability criterion; projects CoM acceleration to support polygon (Vukobratovic, 1972)	**DIRECT: dynamic CoM including inertia**	**NOT PRESENT: models robot inertia, not payload**	ZMP requires dynamic model; Load-Aware A* operates quasi-statically — simpler and faster for slow gaits

Approach Type	Method Description	Stability Model	Payload Support	Key Limitation vs Load-Aware A*
Energy-based stability	Tipover stability index via potential energy landscape; Papadopoulos & Rey (1996)	**DIRECT: energy-based ; continuous metric**	**PARTIAL: some formulations include payload as point mass**	Not integrated into global path search; evaluated post-hoc on candidate paths
Normalised Energy Stability Margin (NESM)	Gravity-normalised stability metric; Hirose et al. (2001)	**DIRECT: gravity-normalised; more physical than SSM**	**PARTIAL: payload approximated as additional gravity term**	Computation-intensive; not embedded in A* search; no O(1) pre-computation

Literature gap — Stream 2: Stability constraint methods achieve rigorous physical modelling but are consistently applied as post-processing validators or hard constraints, not as continuous cost terms in a global path search. Load-Aware A* is the first formulation to embed a payload-aware stability metric as an additive cost in A*, preserving both the algorithm's global optimality and its completeness.

VI-F.3 Stream 3: Sampling-Based Planners with Stability Awareness

Sampling-based planners (RRT, PRM, and their variants) have been applied to legged robot navigation, with several extensions addressing stability. These methods offer probabilistic completeness and can handle high-dimensional configuration spaces, but have specific trade-offs compared to the A* approach.

Method	Planner Type	Stability Integration	Payload Handling	Comparison with Load-Aware A*
RRT (LaValle, 1998) — baseline	Sampling-based tree expansion; probabilistically complete	**NOT PRESENT in base form**	**NOT PRESENT**	Load-Aware A* is resolution-complete (deterministic) and computationally cheaper for 2D grid environments
Stability-RRT (Burridge et al., 1999)	RRT with stability region constraints; prunes unstable samples	**PRESENT: stability checked per sample**	**NOT PRESENT**	Binary pruning vs continuous penalty; Load-Aware A* degrades gracefully when all regions are unstable
RRT* with cost function (Karaman & Frazzoli, 2011)	Asymptotically optimal RRT; arbitrary cost functions possible	**PARTIAL: stability can be encoded as cost; not standard**	**PARTIAL: possible but not demonstrated with payload CoM**	Load-Aware A* achieves deterministic optimality on discrete grids with lower constant-factor overhead than RRT*
PRM with stability edges (Kavraki et al., 1996 variant)	Pre-built roadmap; stability-filtered edges	**PARTIAL: edge filtering by stability metric**	**NOT PRESENT**	Roadmap requires offline construction; Load-Aware A* plans online with S_map computed per scenario

Literature gap — Stream 3: Sampling-based planners with stability extensions either use binary pruning (completeness risk) or require offline roadmap construction (no online adaptability). Load-Aware A* achieves online, deterministic, globally optimal planning with continuous stability integration in O(V log V) time on discrete grids — a better match for the structured indoor/structured outdoor environments where quadruped robots typically operate.

VI-F.4 Stream 4: Learning-Based and Reactive Approaches

Recent work has applied reinforcement learning and neural-network-based planners to legged robot navigation, including stability-aware variants. While these methods show promise for unstructured environments, they present a different set of trade-offs for the payload-carrying scenario studied here.

Approach	Method	Stability Handling	Payload Adaptability	Comparison with Load-Aware A*
Deep RL locomotion (Kumar et al., 2021)	Policy gradient on physical robot; terrain adaptation via proprioceptive feedback	**DIRECT: stability emergent from reward shaping**	**PARTIAL: can adapt to payload if trained with it**	RL requires training per robot/payload config; Load-Aware A* adapts analytically via Eq. 4 with no re-training
Neural A* (Yonetani et al., 2021)	CNN-guided A* with learned traversability	**PARTIAL: traversability can encode stability implicitly**	**NOT PRESENT: no explicit payload CoM model**	Load-Aware A* provides explicit, interpretable stability justification; Neural A* is a black box w.r.t. stability
DWA with stability cost (reactive)	Dynamic Window Approach with stability penalty in velocity space	**PARTIAL: local stability; no global path**	**PARTIAL: payload as inertia parameter**	Load-Aware A* is a global planner; DWA is local/reactive; different scope, potentially complementary

Literature gap — Stream 4: Learning-based methods require per-configuration training and lack interpretable stability guarantees. Load-Aware A* provides analytic, explainable stability routing that adapts to arbitrary payload configurations through the physics of Equation 4, without re-training or pre-computation beyond the per-scenario O(R×C) S_map build.

VI-F.5 Consolidated Position in Literature

The four literature streams above define the gap that Load-Aware A* fills. The following table summarises Load-Aware A*'s position relative to the most relevant representative works:

Property	Standard A*	Stability-RRT	NESM Methods	Neural A*	Load-Aware A* (this work)
Global path optimality	**YES: geometric**	**PARTIAL: probabilistic**	**NOT PRESENT : post-hoc**	**PARTIAL: learned**	**YES: stability-cost optimal**

Property	Standard A*	Stability-RRT	NESM Methods	Neural A*	Load-Aware A* (this work)
Continuous stability cost	**NOT PRESENT**	**NOT PRESENT: binary**	**YES: physical metric**	**PARTIAL: implicit**	**YES: S(n) graded penalty**
Payload CoM modelling	**NOT PRESENT**	**NOT PRESENT**	**PARTIAL: some variants**	**NOT PRESENT**	**YES: explicit Eq. 4**
Admissibility preserved	**YES**	**NOT APPLICABL E**	**NOT APPLICA BLE**	**NOT APPLICA BLE**	**YES: proven (Day 20)**
Online planning (no offline build)	**YES**	**YES**	**PARTIAL**	NO: needs training	**YES: S_map in O(R×C)**
Interpretable stability guarantee	**NOT PRESENT**	**PARTIAL**	**YES**	**NOT PRESENT**	**YES: closed-form S(n), M(n)**
Completeness preserved	**YES**	**PARTIAL: probabilistic**	**NOT PRESENT : binary**	**NOT APPLICA BLE**	**YES: continuous penalty**

VI-G. Novelty Statement and Differentiation

Novelty in an academic contribution must be specific, falsifiable, and differentiated from all identified prior work. The following three novelty claims are made for Load-Aware A*, each grounded in the literature analysis of VI-F and the algorithm design of Section IV. These correspond directly to the N1, N2, N3 novelty statements in Methodology v2.0 (Section III-D) but are here expanded into full academic argument form.

VI-G.1 Novelty Claim N1: First Continuous Payload-Aware Stability Penalty in A*

Prior work integrating stability into A*-based planners has used either terrain-derived traversal costs (which approximate stability indirectly from slope or surface type) or binary feasibility constraints (which exclude cells where a stability threshold is violated). Neither approach models payload mass explicitly or embeds the result as a continuous, additive, globally-optimised cost in the A* evaluation function.

Load-Aware A* introduces $S(n) = \|x\_CoM(n) - x\_c(n)\|_2$ (Equation 5) as a continuous stability penalty derived directly from the robot's physical configuration — specifically, from the payload-mass-weighted Centre of Mass position (Equation 4). This penalty is:

- Continuous and graded: every cell receives a distinct S(n) value proportional to the actual CoM displacement, not a binary safe/unsafe classification
- Physically exact: derived from the mass-weighted CoM model, not approximated from terrain geometry
- Globally optimised: accumulated in g(n) across the entire path, so the algorithm minimises total stability cost rather than avoiding individual dangerous cells
- Admissibility-preserving: positioned in g(n) only, leaving h(n) = Manhattan unchanged and maintaining the A* optimality guarantee

No prior published A* variant for legged robots combines all four of these properties simultaneously. This constitutes a genuine algorithmic novelty.

Novelty N1 — Precise claim: Load-Aware A* is, to the authors' knowledge, the first A* variant to embed a payload-mass-dependent, physically-derived, continuous stability penalty as an additive component of the global cost function while preserving heuristic admissibility.

VI-G.2 Novelty Claim N2: Single-Pass Globally Optimal Stability-Aware Planning

The standard approach to stability-aware path planning in prior literature is a two-phase architecture: (1) find a geometrically optimal or feasible path using a standard planner, then (2) evaluate or improve its stability through post-processing. Representative examples include applying NESM checks to RRT-generated paths (Papadopoulos & Rey, 1996 framework) or filtering footstep sequences after A* produces the cell-level path.

This two-phase approach has a fundamental limitation: there is no guarantee that the geometrically optimal path is also the stability-cost optimal path. Post-processing can improve the chosen path but cannot identify a globally better path that a joint optimisation would have found. Load-Aware A* eliminates the two-phase architecture:

- Stability is embedded in the search objective from the first node expansion: the algorithm never finds a path and then asks whether it is stable enough.
- The returned path is globally optimal with respect to the combined objective $f(n) = g(n) + \alpha \cdot S(n) + h(n)$: no alternative path exists with lower combined cost.
- The result is a single search producing a path that simultaneously satisfies geometric efficiency and stability objectives to the best jointly achievable degree.

Novelty N2 — Precise claim: Load-Aware A* achieves single-pass, globally optimal planning with respect to a stability-aware composite cost function. This is structurally different from two-phase methods that apply stability as a post-processing filter on a geometrically-planned path, and provides a stronger optimality guarantee than any post-processing approach.

VI-G.3 Novelty Claim N3: Empirically Validated Alpha Selection with Backwards Compatibility

The stability weight α in the cost function $f(n) = g(n) + \alpha \cdot S(n) + h(n)$ introduces a design parameter that must be set. Prior work that has used weighted cost functions in A* variants typically selects weights heuristically or through grid search without formal justification, and does not explicitly address the degenerate case where the weight collapses the modified algorithm to its baseline.

Load-Aware A* addresses both issues explicitly:

- $\alpha = 1.0$ is the empirically selected minimum value that produces consistent path divergence at m□ ≥ 5 kg, validated across 10 held-out scenarios covering $\alpha \in \{0.1, 0.5, 1.0, 1.5, 2.0\}$. The selection criterion is operationally meaningful: the minimum stability-effective weight, not an arbitrary value.
- When $\alpha = 0$, Load-Aware A* reduces exactly to Standard A* — producing identical paths, costs, node counts, and timings. This backwards compatibility is formally provable from the cost function and verified experimentally. It means Load-Aware A* is a strict generalisation of Standard A*, not a separate algorithm.

- The backwards compatibility provides a built-in regression test: any correct implementation of Load-Aware A* must pass the α = 0 equivalence check, giving the algorithm a falsifiable internal consistency criterion absent from most modified planners in the literature.

Novelty N3 — Precise claim: The α = 1.0 selection is justified by an operationally meaningful criterion (minimum divergence-producing weight), and the α = 0 degenerate case provides formal backwards compatibility with Standard A*. Together, these properties make Load-Aware A* a principled, verifiable extension rather than an ad-hoc modification.

VI-G.4 Summary Novelty Table

Novelty Claim	Prior State of Art	This Work's Contribution	Evidence
N1: Continuous payload CoM penalty in A*	**NOT PRESENT: binary constraints or terrain proxies only**	**NOVEL: $S(n) = \lVert x\_CoM - x\_c \rVert_2$ as additive cost; continuous; payload-exact; admissibility-preserved**	Eq. 4, 5, 10, 11; admissibility proof Day 20
N2: Single-pass globally optimal stability planning	**NOT PRESENT: two-phase (plan then validate) is standard**	**NOVEL: stability embedded in search objective; globally optimal w.r.t. composite cost in one pass**	Algorithm design Section IV; Day 18 analysis
N3: Empirically validated α with backwards compatibility	**NOT PRESENT: weight selection ad-hoc; no degenerate-case analysis**	**NOVEL: α = min divergence-producing weight; α=0 ≡ Standard A* proven and verified**	Day 7 tuning; AMD-02 regression test

VI-H. Extended Limitations Analysis

A rigorous limitations section distinguishes between three categories of constraint: inherent limitations of the algorithm as formulated, scope boundaries that define the work's operating domain, and engineering limitations that could be addressed in future work. Each limitation below is classified, quantified where possible, and paired with a mitigation or extension path. This structure ensures the limitations section is constructive rather than merely defensive.

VI-H.1 Limitation L1: Quasi-Static Assumption

Dimension	Detail
Category	Inherent — baked into the physical model (Equations 4–6)
Description	The stability model assumes the robot is in quasi-static equilibrium at each cell. S(n) and M(n) are computed from static CoM positions without accounting for inertial forces during locomotion.

Dimension	Detail
Impact	At walking speeds above approximately 0.3 m/s, dynamic CoM shifts due to acceleration and deceleration become significant. The planned path may under-estimate instability in cells traversed at high speed.
Quantification	For m_p = 9 kg at v = 0.5 m/s, the dynamic CoM shift from inertia is approximately 0.03–0.05 m — comparable in magnitude to the safety threshold of 0.05 m. At this speed, quasi-static planning may produce a technically invalid safety guarantee.
Mitigation	Apply a speed-dependent safety margin: if the robot traverses at speed v, require $M(n) > 0.05 + k \cdot v^2$ where k is a robot-specific inertia constant. This is a conservative patch, not a full dynamic model.
Future work	Replace Eq. 5 with a Zero-Moment Point (ZMP) based stability cost that incorporates velocity and acceleration. This would extend Load-Aware A* from quasi-static to dynamic locomotion at the cost of a more complex pre-computation step.

VI-H.2 Limitation L2: 2D Grid Abstraction

Dimension	Detail
Category	Scope boundary — defines the work's applicable environment class
Description	The environment is a discrete 2D occupancy grid (R=C=10; cell size 1.0 m). The terrain within each cell is assumed flat and level. Slope, surface texture, and 3D geometry are not modelled.
Impact	On inclined terrain, gravity projects onto the slope normal, shifting the effective CoM toward the downhill edge of the support polygon. This creates stability hazards not captured by the current S(n) model, which computes CoM displacement on a flat-plane assumption.
Quantification	On a 10° slope with m_p = 7 kg, the effective CoM displacement due to gravity component is approximately 0.06 m — enough to push M(n) below the 0.05 m threshold even for cells that appear stable on a flat grid.
Mitigation	The current formulation is valid for flat or near-flat indoor environments (e.g. warehouses, factories) which are the primary deployment contexts for payload-carrying quadrupeds.
Future work	Extend S_map computation to use a 3D height map: compute S(n) with the effective gravity vector projected onto the terrain normal at each cell. This requires a 3D CoM model but preserves the S_map pre-computation architecture and O(1) lookup during search.

VI-H.3 Limitation L3: Single Rigid Payload

Dimension	Detail
Category	Scope boundary — defines payload type applicability
Description	The CoM model (Equation 4) assumes a single rigid payload at a fixed position x_payload. The payload mass and position are constant throughout the path.
Impact	Shifting payloads (liquid containers, pendulum-suspended loads, flexible payloads) have a CoM that varies with robot orientation, acceleration, and vibration. The current S_map would be computed for a fixed payload position and would not reflect the actual dynamic CoM during traversal.

Dimension	Detail
Quantification	For a liquid payload with 20% fluid shift, x_payload may vary by ±0.1 m during motion. At m□ = 7 kg, this translates to a CoM variation of ≈0.04 m — nearly equal to the safety threshold.
Mitigation	For rigid payloads (the most common industrial case), the limitation does not apply. The paper should state explicitly that the model assumes rigid payload with fixed x_payload.
Future work	Model payload CoM as a bounded uncertainty set: compute S(n) for the worst-case payload position within the uncertainty bounds. This converts the deterministic penalty to a robust stability cost, compatible with the existing A* framework.

VI-H.4 Limitation L4: Static Foot Contact Polygon

Dimension	Detail
Category	Inherent — tied to the robot model (Equation 3)
Description	The support polygon is computed from four fixed foot positions (Equation 3) assuming all four feet are simultaneously in contact. During dynamic gaits (trot, gallop), two or more feet leave the ground at once, reducing the support polygon to a triangle or line.
Impact	During a trot gait, the effective support polygon alternates between two diagonal pairs of feet. The resulting polygon is substantially smaller than the four-contact polygon, making M(n) values computed from Equation 6 systematically over-optimistic.
Quantification	For hs = 0.4 m, the four-contact polygon has area 0.64 m^2. A diagonal two-contact trot polygon has area 0 m^2 (degenerate line) — M(n) = 0 by definition during the float phase. The current model cannot capture this.
Mitigation	The four-contact model is accurate for standing and slow-walking gaits (creep gait), which are the appropriate gaits for payload carrying. The paper's applicability statement should specify 'slow-walking (creep gait) or standing manoeuvre'.
Future work	Extend the foot contact model to enumerate all gait phases and compute a worst-case M(n) across phases. This requires a gait model as input but preserves the S_map pre-computation structure.

VI-H.5 Limitation L5: Grid Resolution and Discretisation Error

Dimension	Detail
Category	Engineering — addressable by increasing resolution
Description	The 10×10 grid with 1.0 m cell size means path planning operates at 1-metre spatial resolution. The robot's actual foot placement within a cell is not considered; the entire cell is assigned a single S(n) value based on the cell-centre CoM position.
Impact	Near payload positions where the S(n) gradient is steep, adjacent cells may have substantially different actual CoM displacements depending on where within the cell the robot steps. The cell-level S(n) is an average that may under-estimate instability at cell edges.

Dimension	Detail
Quantification	At 1.0 m cell size, the sub-cell S(n) variation can be up to 0.4 m (half the foot span) — significant relative to the 0.05 m safety threshold. Finer grid resolution reduces this error proportionally.
Mitigation	For the 10×10 benchmark grid, this is acceptable as a research study demonstrating the algorithm's principle. Real deployments on 0.2 m or 0.5 m grids would reduce discretisation error by 5× or 2× respectively.
Future work	Apply sub-cell S(n) computation: sample S at multiple foot positions within each cell and use the maximum (worst-case) as the cell penalty. Alternatively, use a continuous-space S function with lazy evaluation during A* expansion.

VI-H.6 Limitations Summary and Research Roadmap

Limitation	Category	Impact Level	Current Mitigation	Extension Path
L1: Quasi-static assumption	Inherent	**HIGH at v > 0.3 m/s; LOW for slow gaits**	Speed-dependent margin addition	ZMP-based dynamic cost function
L2: 2D grid / flat terrain	Scope boundary	**HIGH on slopes > 5°; LOW indoors**	Valid for flat-floor environments	3D height map S_map extension
L3: Single rigid payload	Scope boundary	**HIGH for liquids; LOW for rigid loads**	State rigid payload assumption	Bounded uncertainty CoM model
L4: Static foot polygon	Inherent	**HIGH for trot gait; LOW for creep gait**	Restrict applicability to slow/stand gaits	Worst-case gait-phase foot model
L5: Grid discretisation	Engineering	**MODERATE at 1.0 m; LOW at 0.2 m**	Acceptable for benchmark; note in experiments	Sub-cell S(n) sampling

Limitations framing for paper: Present these five limitations as a Research Roadmap, not a list of failures. Each limitation is a well-defined open problem with a clear extension path, demonstrating that Load-Aware A* is a foundation for a family of increasingly capable planners rather than a closed, narrow contribution.

PART B — MEMBER NOTES & DISCUSSION COMPLETION CHECKLIST

2. Section VI Completion Status

With today's additions (VI-F, VI-G, VI-H), Section VI is fully drafted. The table below records the completion status of all eight Discussion sub-sections across Days 25–26.

Sub-section	Title	Drafted	Day
VI-A	Why Load-Aware A* performs better than Standard A*	**YES ✔**	Day 25

Sub-section	Title	Drafted	Day
VI-B	Theoretical justification (admissibility + physical mechanics)	**YES ✔**	Day 25
VI-C	Stability benefits — quantitative analysis	**YES ✔**	Day 25
VI-D	Payload scalability and parameter robustness	**YES ✔**	Day 25
VI-E	Limitations — scope overview (first introduction)	**YES ✔**	Day 25
VI-F	Comparison with prior literature (4 streams)	**YES ✔**	Day 26
VI-G	Novelty statement and differentiation (N1, N2, N3)	**YES ✔**	Day 26
VI-H	Extended limitations analysis (L1–L5 with research roadmap)	**YES ✔**	Day 26

Note on VI-E vs VI-H: Section VI-E (Day 25) is a brief scope-overview limitations statement intended to appear near the end of the performance discussion, before the literature comparison. Section VI-H (Day 26) is the full extended limitations analysis with mitigation paths, intended as the penultimate Discussion sub-section before the concluding summary. Both should appear in the final paper; VI-E provides a bridge from results to literature context, VI-H provides the deep scholarly engagement reviewers expect.

3. Discussion — Related Work Section Note

Depending on the target journal or conference format, the literature review in VI-F may be repositioned as a standalone Section II (Related Work) before the Methodology, rather than inside the Discussion. The content is identical; only the placement changes. The following guidance applies:

- IEEE Transactions format: Related Work typically appears as Section II before Methodology. Move VI-F to Section II if this format is used.
- ICRA / IROS conference format: Related Work is often brief (0.5–1 column) and appears in Section I (Introduction) as a literature paragraph. Condense VI-F to 3–4 representative citations per stream if space is limited.
- If VI-F stays in Discussion: Rename it ‘Comparison with Prior Work’ and ensure it directly references the results in Section V to maintain Discussion-section coherence.